

Computational Physics: Materials Science

Exercise 5

NVT ensemble and isochoric heat capacity

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NVT Ensemble and Isochoric Heat Capacity of Na₂₇

This exercise extends the Ex. 4 molecular-dynamics code for the 27-atom sodium cluster (Morse potential, shifted cutoff at $r_{\text{cut}} = 2.5 r_0$, minimum-image periodic boundaries) to the NVT ensemble. The goal is to estimate the isochoric heat capacity $C_V(T)$ by two independent estimators and, crucially, to identify the temperature ranges and thermostat choices under which each estimator does, or does not, return a physically meaningful answer. The Atomic Simulation Environment (ASE) drives the dynamics; the numba force kernel from Ex. 4 is reused unchanged and wrapped as a custom ASE `Calculator`. An AI assistant (Claude) helped with the ASE integration and plot generation; the physical interpretation is my own.

Implementation

Heat-capacity estimators

Method 1 uses the canonical-ensemble fluctuation–dissipation relation. It is valid only if (i) the sampled distribution is canonical *and* (ii) the single production trajectory is long enough to converge the energy variance:

$$C_V(T) = \frac{\langle (\delta E_{\text{tot}})^2 \rangle}{k_B T^2} = \frac{\langle E_{\text{tot}}^2 \rangle - \langle E_{\text{tot}} \rangle^2}{k_B T^2}. \quad (1)$$

Method 2 approximates C_V by a centred finite difference of the mean total energy across two independent runs at $T \pm \Delta T$:

$$C_V(T) \approx \frac{\langle E_{\text{tot}}(T + \Delta T) \rangle - \langle E_{\text{tot}}(T - \Delta T) \rangle}{\Delta T_{\text{meas}}}. \quad (2)$$

This is the sheet’s finite-difference formula with one deliberate refinement: the denominator uses the *measured* mean-temperature difference ΔT_{meas} rather than the nominal $2\Delta T$. The two coincide when the thermostat lands exactly on its target, but diverge badly when it does not — a non-negligible effect for Langevin at high T (Sec. (f)). Using ΔT_{meas} keeps Eq. (2) an unbiased estimate of $d\langle E \rangle/dT$ between the two points actually sampled. The two estimators are thermodynamically identical, $d\langle E \rangle/dT = \text{Var}(E)/(k_B T^2)$, so wherever both are well behaved they must agree; where they disagree, the disagreement is itself diagnostic.

Code stack

The potential, unit system, shifted cutoff and numba force kernel are inherited unchanged from Ex. 4. New to Ex. 5:

- **ASE Calculator wrapper.** A subclass of `ase.calculators.Calculator` reads positions from an `ase.Atoms` object, calls the numba Morse kernel, and writes energy and forces into the `results` dictionary. This is the minimum needed to make ASE’s MD drivers use the exact Ex. 4 potential rather than ASE’s built-in `MorsePotential`, whose functional form and cutoff handling differ.
- **Langevin thermostat.** `ase.md.langevin.Langevin` with friction $\gamma = 0.01 \text{ fs}^{-1}$ (relaxation time $\approx 100 \text{ fs}$, matched to the Berendsen coupling time). `fixcm=False`;



the alternative `fixcm=True` is the deprecated ASE default, and ASE itself warns that it does *not* strictly sample the canonical distribution for small systems.

- **Berendsen thermostat.** `ase.md.nvtberendsen.NVTBerendsen` with $\tau_T = 100$ fs, identical to the Ex. 4 setting.

In Method 1 the variance is divided by the *target* temperature T_0 , not the measured mean $\langle T \rangle$. This is the conventional choice; using $\langle T \rangle$ instead would shift, for example, the 5000 K Langevin point from 1.60 to $\approx 1.47 Nk_B$ (even closer to the ideal-gas value), because $\langle T \rangle = 5213$ K there.

Simulation parameters

N	ρ [atoms/nm ³]	L [Å]	Δt [fs]	N_{equil}	N_{prod}	γ [fs ⁻¹]	τ_T [fs]
27	0.5	37.80	1	12000	5000	0.01	100

Table 1: Simulation parameters used in all production runs. The equilibration window is deliberately longer than the production window ($12000 > 5000$) because at the lowest temperatures the simple-cubic seed relaxes only slowly into a compact configuration; if production starts before that relaxation settles, the Method-1 variance picks up the drift rather than the equilibrium fluctuation (see Sec. (d)).

Analysis

(a) Berendsen and Langevin at $T_0 = 300$ K

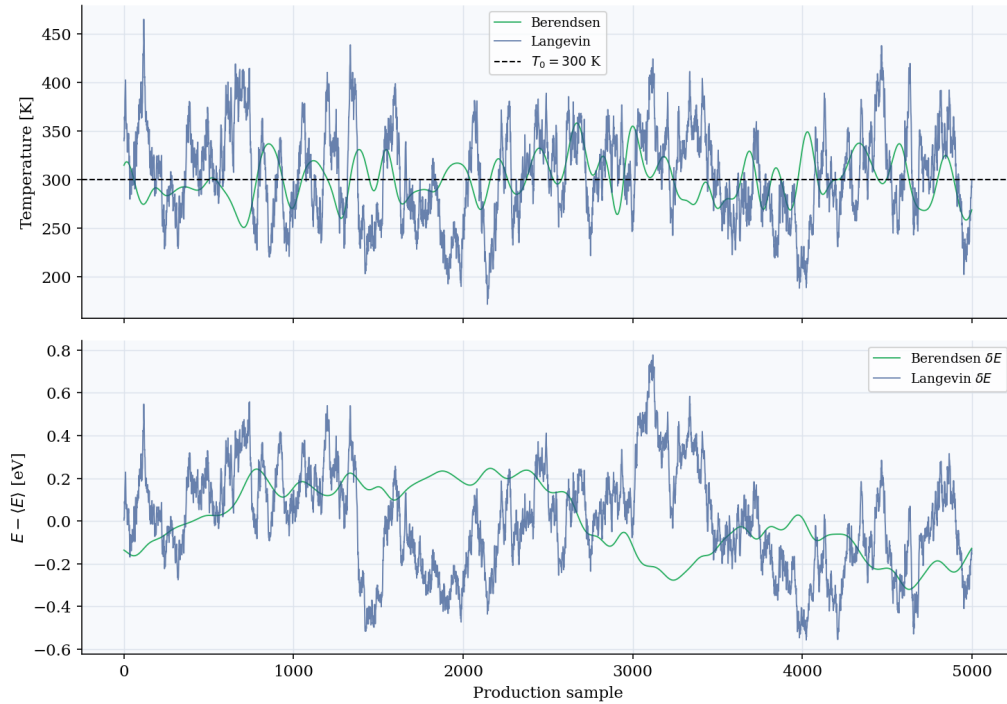


Figure 1: Berendsen (green) and Langevin (blue) at $T_0 = 300$ K, both sampled every step over 5000 production steps after 12000 equilibration steps. Top: instantaneous temperature. Bottom: total energy minus its mean. Both thermostats hold $\langle T \rangle \approx T_0$; the amplitude of the fluctuations differs sharply.



Both thermostats control the mean temperature successfully: $\langle T \rangle_{\text{Lang}} \approx 304 \text{ K}$ and $\langle T \rangle_{\text{Ber}} \approx 300 \text{ K}$. They differ in the *spread* of the instantaneous temperature:

- Langevin: $\sigma_T \approx 47.6 \text{ K}$. The canonical prediction for a classical system of $3N$ degrees of freedom is $\sigma_T/T_0 = \sqrt{2/(3N)} \approx 15.7\%$, i.e. $\sigma_T \approx 47 \text{ K}$. The match is excellent, confirming that Langevin correctly samples the canonical ensemble.
- Berendsen: $\sigma_T \approx 21.8 \text{ K}$, well below the canonical value. This is the expected consequence of velocity-rescaling: scaling all velocities every step by $\lambda = \sqrt{1 + (\Delta t/\tau_T)(T_0/T - 1)}$ damps any excursion away from T_0 and therefore suppresses the kinetic and total-energy fluctuations.

The same contrast appears in the lower panel of Fig. 1: Langevin’s $\sigma_E \approx 0.24 \text{ eV}$, while Berendsen’s stays near 0.16 eV and is visibly smoothed in time.

Conclusion: both thermostats *work* in the weak sense of stabilising $\langle T \rangle$, but only Langevin samples the canonical distribution. Since Eq. (1) measures a property of that distribution, Method-1 results obtained with Berendsen are expected to systematically misrepresent C_V — not because the system is different, but because the sampled ensemble is not the one Eq. (1) assumes.

(b) Method 1 at $T = 5000 \text{ K}$

At 5000 K the thermal energy $k_B T \approx 0.43 \text{ eV}$ exceeds the Morse well depth $D_{\text{Na}} = 0.0633 \text{ eV}$ by a factor of ~ 7 . The cluster cannot remain bound and the 27 atoms disperse into a dilute vapour inside the periodic box. This is consistent with the 5000 K trace in Fig. 3, which sits around $E_{\text{tot}} \approx +17 \text{ eV}$ — the ideal-gas kinetic energy at this temperature is $\frac{3}{2} N k_B T \approx 17.4 \text{ eV}$, so the potential energy contribution is small. The relevant reference is therefore the translational ideal gas, $C_V^{\text{IG}} = \frac{3}{2} N k_B$.

Thermostat	$\langle T \rangle [\text{K}]$	$C_V/(N k_B)$ (Method 1)	vs. ideal gas
Langevin	5213	1.596	within $\sim 7\%$ of ideal gas
Berendsen	4999	0.005	≈ 0 , fluctuations destroyed
ideal gas	—	1.500	—

Table 2: Method-1 heat capacity at 5000 K (independent run, distinct random seed from the sweep in Sec. (c)). Langevin’s measured mean temperature is 5213 K rather than 5000 K , illustrating the loose temperature control at high T ; despite this offset the Method-1 result is close to the ideal-gas reference. Berendsen lands at 4999 K and returns essentially zero, because its sampling damps precisely the fluctuations that Eq. (1) integrates over.

The 5000 K point is the cleanest validation. With no remaining cohesive potential energy to fluctuate, Langevin’s $\text{Var}(E)/(k_B T^2)$ collapses to (within noise) the pure translational contribution $\frac{3}{2} N k_B$, while Berendsen’s collapses to almost zero because its sampling damps precisely the fluctuations that Eq. (1) integrates over. (Note that an independent sweep run at the same target gives $1.29 N k_B$, Table 3; the 1.29 – 1.60 spread between two independent runs is itself within the $\sim 20\%$ statistical noise floor estimated in Sec. (c).)



(c)+(e) Temperature sweep, Method 1

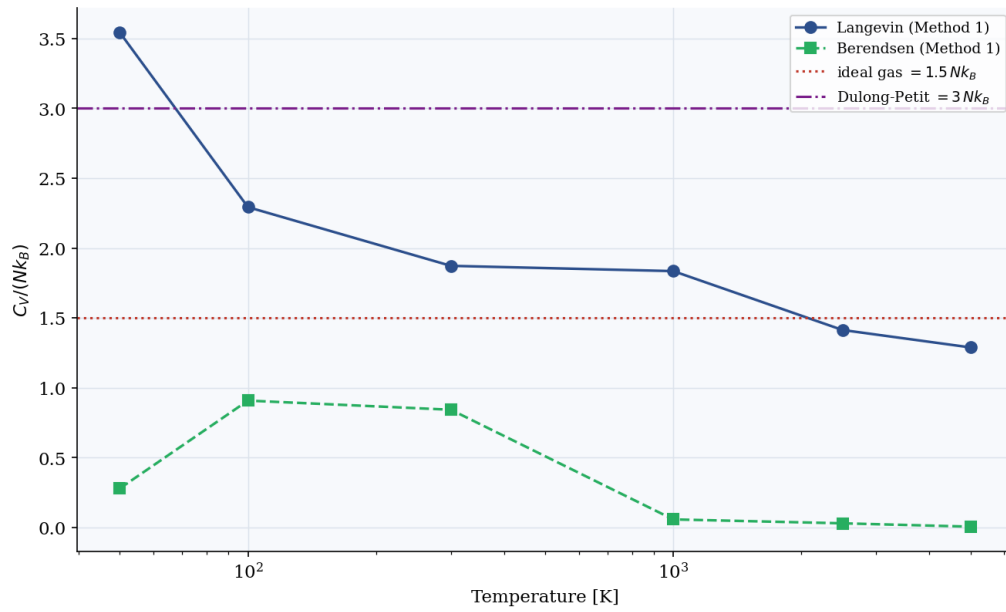


Figure 2: Heat capacity per atom from Method 1 across $T \in \{50, 100, 300, 1000, 2500, 5000\}$ K, for Langevin (blue, circles) and Berendsen (green, squares). Reference lines: ideal gas $1.5 Nk_B$ (red dotted) and bulk Dulong–Petit $3 Nk_B$ (purple dash-dotted). The relevant ceiling for a *finite* 27-atom cluster is slightly lower, $\approx 2.89 Nk_B$ (Sec. (d)). Temperature axis is logarithmic.

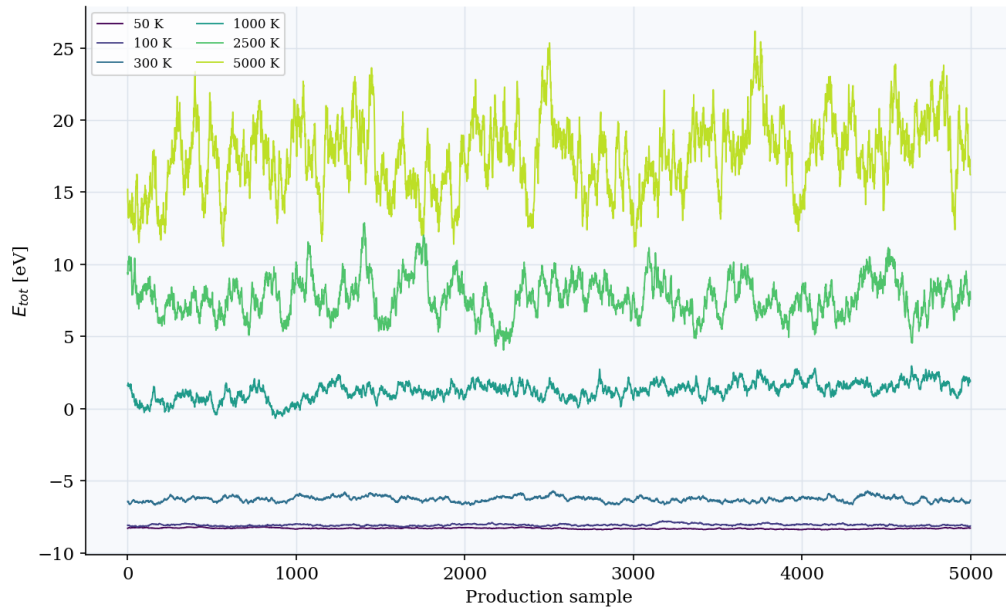


Figure 3: Total energy along the production phase of the six Langevin sweep runs. Each trace is stationary about its mean over the 5000-step production window. The vertical ordering is the expected one: cold runs sit at $E_{\text{tot}} \approx -8$ eV (bound cluster, mostly potential energy); the hottest run at $\approx +17$ eV (free atoms, mostly kinetic energy, $\frac{3}{2}Nk_B T \approx 17.4$ eV at 5000 K). The jump in mean energy between 300 K and 1000 K shows where the cluster begins to fragment.



The cold-cluster plateau near -8 eV is not in tension with the single-bond well depth $D_{\text{Na}} = 0.0633\text{ eV}$: it is the sum over *all* pair interactions inside the $13.34\text{ \AA}$ cutoff. A compact 27-atom cluster has of order fifty nearest-neighbour contacts near the well bottom, plus many weaker outer-shell pairs, so the total binding energy is two orders of magnitude larger than D_{Na} .

Reading the Langevin curve in Fig. 2 from low to high temperature. The single most important point for interpreting this curve is that Method 1 has *two* failure modes, and both appear here (see Sec. (d)): a too-short single trajectory can either over- or under-estimate the variance.

- $T = 50\text{ K}$: $C_V/(Nk_B) \approx 3.54$, *above* both the bulk Dulong–Petit value and the cluster ceiling of 2.89. This is not a physical C_V : at 50 K the simple-cubic seed is still slowly relaxing during production, and that residual drift inflates $\text{Var}(E)$ in Eq. (1) (Sec. (d)).
- $T = 100\text{ K}$: $C_V/(Nk_B) \approx 2.29$, within the $\sim 20\%$ statistical noise of the cluster-harmonic ceiling $2.89 Nk_B$ (Sec. (d)). Here the cluster is a well-behaved harmonic solid: the energy fluctuations are dominated by fast vibrational modes that the short run samples adequately, so Method 1 is reliable.
- $T = 300\text{ K}$: $C_V/(Nk_B) \approx 1.87$. **This value should not be read as the physical C_V .** Both the Ovito snapshot (Fig. 4) and Method 2 ($\approx 3.4\text{--}3.6 Nk_B$, Sec. (f)) show a *bound, solid-like* cluster at 300 K. The low Method-1 value is a convergence limitation of the fluctuation estimator: a single 5 ps trajectory does not explore the configurational rearrangements that begin to open up near the crossover, so it under-counts $\text{Var}(E)$. Method 2, which compares two independent equilibrated means, recovers the missing heat capacity (Sec. (f)).
- $T = 1000\text{ K}$: $C_V/(Nk_B) \approx 1.83$. Note that the mean total energy has jumped from -8 to $\approx +1\text{ eV}$ (Fig. 3), so the cluster is now clearly fragmenting — yet the Method-1 value is barely different from 300 K. This near-equality is itself the tell-tale sign that the 300 K Method-1 point is anomalously low: a bound solid should have a *larger* C_V than a fragmenting cluster, not the same.
- $T = 2500\text{ K}$: $C_V/(Nk_B) \approx 1.41$, within statistical noise of the ideal-gas line. Mixing is fast in the gas-like regime, so Method 1 is reliable again.
- $T = 5000\text{ K}$: $C_V/(Nk_B) \approx 1.29$, see Sec. (b); the part-(b) run at the same target gave 1.60, the spread illustrating the noise floor below.

Statistical noise floor. With $N_{\text{prod}} = 5000$ samples and a Langevin velocity autocorrelation time $\tau_{\text{corr}} \approx 1/\gamma = 100$ steps, each curve point is built from ~ 50 effectively independent samples. The relative error on a variance estimated from N_{eff} independent samples is $\sqrt{2/N_{\text{eff}}} \approx 20\%$. This sets the floor on the *vibrational* part; the configurational under-sampling at 300 K (above) is a separate, larger effect not captured by this estimate.

The Berendsen curve (green) sits below the Langevin curve at every temperature. At high T ($\geq 1000\text{ K}$) it collapses to ≈ 0 , mirroring Sec. (b). The intermediate values should not be read as *measurements* of C_V : they are an artefact of the Berendsen sampling not being canonical, and their magnitude depends on τ_T rather than on the thermodynamics of the cluster.



(d) Comparison to ideal gas and Dulong–Petit; why some points fail

The classical equipartition reference values are:

- translational ideal gas: $C_V = \frac{3}{2} Nk_B$;
- bulk 3D harmonic solid (Dulong–Petit): $C_V = 3 Nk_B$ — the purple line in Fig. 2;
- *finite-cluster* harmonic ceiling: a free 27-atom cluster has 6 zero-frequency modes (3 translation + 3 rotation) that carry kinetic but no potential energy. Counting quadratic terms gives $C_V = (3N - 3) k_B \approx 2.89 Nk_B$, slightly below the bulk value. This, not $3 Nk_B$, is the correct harmonic target for our system.

Three reasons the Method-1 curve does not interpolate cleanly between the ideal-gas and harmonic limits:

(i) Classical MD has no quantum freeze-out. Real materials follow the Debye law $C_V \propto T^3$ as $T \rightarrow 0$, because vibrational modes with $\hbar\omega \gtrsim k_B T$ are frozen out. Classical MD has no $\hbar$ and predicts the full harmonic value all the way to $T \rightarrow 0$. Our 100 K value of $\approx 2.3 Nk_B$ is the classical limit; the experimental Na heat capacity at this temperature is much smaller.

(ii) Method 1 is sensitive to the quality of a single trajectory. The fluctuation estimator assumes the production run has converged the equilibrium variance. Two opposite failures appear in our sweep: at 50 K ($3.54 Nk_B$) residual relaxation of the non-equilibrium seed *inflates* $\text{Var}(E)$; at 300 K ($1.87 Nk_B$) a single short run *under-explores* the configurational space that opens near the melting/evaporation crossover, *deflating* $\text{Var}(E)$. Neither is a property of the cluster. Method 2 needs only mean energies and is immune to both; it returns the reliable solid-like value $\approx 3.4\text{--}3.6 Nk_B$ at 300 K (Sec. (f)).

(iii) Berendsen everywhere. As shown in Sec. (a), the Berendsen distribution is not canonical. Equation (1) then simply does not apply, and the Berendsen curve in Fig. 2 is a demonstration that Method 1 fails for this thermostat.

Finally, no sharp melting peak is resolved. For a 27-atom cluster the solid–liquid–vapour crossover is broad (a small finite system does not exhibit a true phase transition), and our sweep leaves a wide gap between 300 K (bound) and 1000 K (fragmenting) — exactly where the cluster’s depressed melting lies. A dedicated scan in the $\sim 400\text{--}700$ K window would be needed to localise the crossover; with the present coarse sampling and $\sim 20\%$ noise per point, no peak can be placed.

(f) Method 2 at $T = 300$ K and 5000 K

T [K]	Thermostat	M1 [Nk_B]	M2 [Nk_B]	ΔT_{meas} [K]	note
300	Langevin	1.87	3.57	191.3	M1 under-sampled; M2 is reliable
	Berendsen	0.84	3.41	198.6	M1 non-canonical; M2 recovers it
5000	Langevin	1.29	2.83	47.5	M2 unreliable: $\Delta T_{\text{meas}} \ll 2\Delta T$
	Berendsen	0.01	1.47	200.2	M2 close to ideal gas ($1.5 Nk_B$)

Table 3: Comparison of Method 1 (energy fluctuations) and Method 2 (finite difference of $\langle E_{\text{tot}} \rangle$, nominal $\Delta T = 100$ K). ΔT_{meas} is the *measured* mean-temperature difference between the $T \pm \Delta T$ runs, used in the denominator of Eq. (2). The Method-1 values are the sweep values of Sec. (c).

Four observations from Table 3:



- **Method 2 at 300 K (both thermostats).** Both give $C_V \approx 3.4\text{--}3.6 Nk_B$, just above the finite-cluster harmonic ceiling of $2.89 Nk_B$. This is the expected signature of a bound solid entering its anharmonic / pre-melting regime, and is consistent with the compact cluster seen in Ovito (Fig. 4). Method 2 needs only the *mean* energy, which both thermostats reproduce correctly.
- **The Langevin M1/M2 gap exposes Method-1 under-sampling.** Langevin samples the canonical ensemble correctly, so Eqs. (1) and (2) are the *same* quantity and ought to agree. The fact that Langevin Method 1 (1.87) sits at roughly half of Langevin Method 2 (3.57) is therefore not a physical effect but a clear statement that the single 300 K trajectory did not converge the energy variance. Method 2, comparing two independent equilibrated means 191.3 K apart, captures the configurational heat the single run misses. The reliable 300 K value is $\approx 3.5 Nk_B$.
- **Berendsen at 300 K fails for a different reason.** Method 1 with Berendsen returns only $0.84 Nk_B$ because the Berendsen distribution is not canonical and Eq. (1) does not apply at all. Method 2 with Berendsen returns $3.41 Nk_B$, consistent with the Langevin Method 2 result — the failure was in the *sampling*, not the system.
- **At 5000 K, the reliable estimates agree with the ideal gas.** Berendsen Method 2 gives $1.47 Nk_B$, within 2% of $1.5 Nk_B$, because its tight temperature control lands the two bracketing runs almost exactly 200 K apart ($\Delta T_{\text{meas}} = 200.2 \text{ K}$). Langevin Method 2 is unreliable: loose control collapsed the bracket to $\Delta T_{\text{meas}} = 47.5 \text{ K}$, and dividing $\Delta \langle E \rangle$ by such a small, noisy denominator yields $2.83 Nk_B$ — not physically meaningful.

The compact statement: *Method 1 needs both canonical sampling and a converged single-trajectory variance; Method 2 needs only correct mean energies.* Method 1 fails in two distinct ways here — non-canonical sampling (Berendsen, all T) and an unconverged variance (Langevin’s 50 K drift and 300 K under-exploration) — whereas Method 2 is robust wherever $\langle T \rangle$ is well controlled, most reliably with Berendsen.

(c) Ovito visualisation of the trajectories

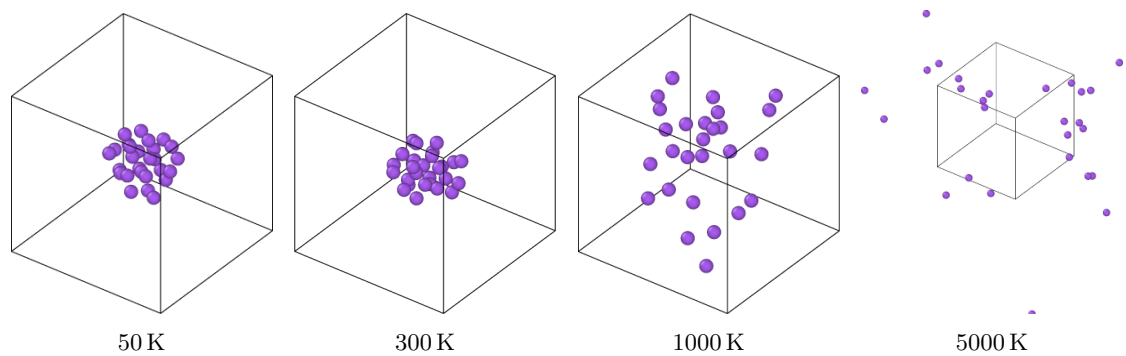


Figure 4: Snapshots from Ovito of the Langevin trajectories at four representative temperatures, taken from the end of the production window. At 50 K and 300 K the cluster retains a compact, bound arrangement — direct visual confirmation that the 300 K state is solid-like, in line with the Method-2 value of $\approx 3.5 Nk_B$ and against the under-sampled Method-1 value. At 1000 K the cluster is visibly expanded and some atoms begin to leave the central core, consistent with the onset of fragmentation seen in Fig. 3. At 5000 K the 27 atoms are fully dispersed throughout the periodic box, forming a dilute vapour — the ideal-gas limit of Secs. (b) and (d).



Summary

- **Thermostats.** Both Langevin and Berendsen hold $\langle T \rangle$ at T_0 . Only Langevin reproduces the canonical temperature fluctuation $\sigma_T \approx T_0 \sqrt{2/(3N)} \approx 47$ K (measured: 47.6 K); Berendsen suppresses fluctuations to $\sigma_T \approx 22$ K.
- **Method 1 at 5000 K.** Langevin returns $C_V \approx 1.60 Nk_B$ (an independent sweep run gives 1.29; the spread is the $\sim 20\%$ noise floor), consistent with the translational ideal gas ($1.5 Nk_B$). Berendsen returns essentially zero, because its sampling damps precisely the fluctuations Method 1 measures.
- **Method 1 sweep.** Method 1 is reliable where mixing is fast: near the ideal-gas value for $T \geq 1000$ K, and near the cluster-harmonic ceiling ($\approx 2.89 Nk_B$) at 100 K ($2.29 Nk_B$). The 50 K (3.54, drift-inflated) and 300 K (1.87, variance-unconverged) points are *estimator artefacts*, not physical C_V values.
- **Method 2.** Robust because it needs only mean energies. At 300 K both thermostats give $\approx 3.4\text{--}3.6 Nk_B$ — the reliable solid-like value that Method 1 misses. At 5000 K Berendsen gives $1.47 Nk_B$ (close to ideal gas), while Langevin is unreliable ($\Delta T_{\text{meas}} = 47.5$ K $\ll 200$ K).
- **Physical picture of the cluster.** Bound and solid-like up to ~ 300 K (Ovito + Method 2), fragmenting by 1000 K (energy jump in Fig. 3), and a fully dispersed ideal gas by 5000 K.
- **Limitations.** Classical MD cannot capture quantum freeze-out at low T . The Method-1 vibrational noise floor is $\sim 20\%$, and its configurational under-sampling near the crossover is larger still. The finite-cluster harmonic ceiling is $2.89 Nk_B$, not $3 Nk_B$. No sharp melting peak is resolved: for 27 atoms the transition is a broad crossover, and the present sweep does not sample the $\sim 400\text{--}700$ K window where it lies.

